

PC 505 AI	OPERATING SYSTEMS				
Prerequisites	Programming in C and Data Structures	L	T	P	C
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Evaluation	CIE	40 Marks	SEE		60 Marks

Course Objectives

1	To introduce the concepts of OS structure and process synchronization
2	To study different memory management strategies
3	To familiarize the implementation of file system
4	To understand the principles of system security and protection
5	To discuss the design principles and structure of Windows 7 and Linux

Course Outcomes

On completion of this course, the student will be able to

CO1	Evaluate various process scheduling algorithms
CO2	Analyze the steps in address translation and different page replacement strategies
CO3	Compare file allocation methods and choose appropriate allocation strategies for a file.
CO4	Apply the appropriate mechanisms to control access to resources

UNIT – I

CP8-PPT

Introduction to Operating Systems: OS structure and strategies, Process concepts, Multithreaded Programming, Process scheduling, Process synchronization, Deadlocks.

UNIT – II

chp-7 & chp-8.2, 8.6 (CP9&10-PPT)

Memory management strategies with example architectures: Swapping, Contiguous allocation, Paging, Segmentation, Segmentation with paging, Virtual memory management: Demand paging, Page replacement, Thrashing.

UNIT – III

File system interface: File concepts, Access methods and protection. File system implementation: File system structure, Allocation methods, Directory implementation of file systems, Mass storage structures, I/O systems.

UNIT – IV

CP17-PPT

System Protection: Principles and Domain, Access Matrix and implementation, Access control and access rights, Capability based systems, Language based Protection.

System Security: Problem, Program threats, cryptography, user authentication, implementing security defenses, Firewalling, Computer security Classification.

UNIT – V

Case Studies: The Linux System–Design principles, Kernel modules, Process management, Scheduling, Memory management, File systems, Input and Output, Inter process communication. Windows 7 –Design principles, System components, Terminal services and fast user switching File systems, Networking, Programmer Interface.